

Question

Write a Java program to convert a Roman numeral into an integer.

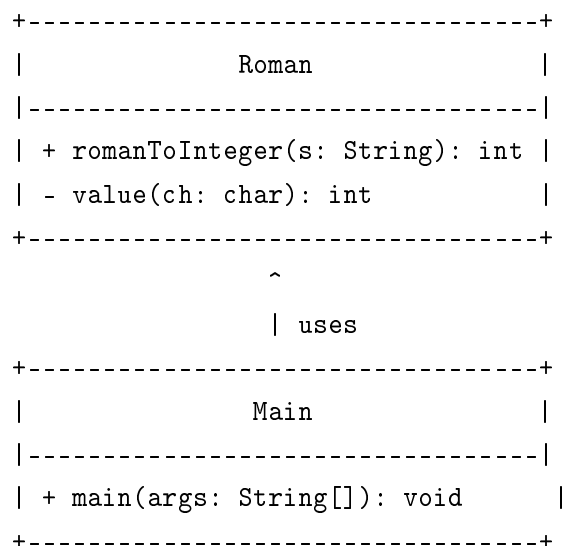
Steps for Execution

1. Save the first code as `Roman.java` in the `number` folder.
2. Save the second code as `Main.java` in the parent folder.
3. Compile both files and run `Main`.
4. Enter a Roman numeral when prompted.

Algorithm

1. Read the Roman numeral from the user.
2. Find the integer value of each Roman symbol.
3. Subtract a symbol when its value is less than the next symbol.
4. Otherwise, add its value to the total.
5. Display the resulting integer.

UML Class Diagram



CODE: SRC

Roman.java

```

package number;

public class Roman {

    public static int romanToInteger(String s) {
        int total = 0;

        for (int i = 0; i < s.length(); i++) {
            int current = value(s.charAt(i));

            if (i + 1 < s.length() && current < value(s.charAt(i + 1))) {
                total -= current;
            } else {
                total += current;
            }
        }

        return total;
    }

    private static int value(char ch) {
        switch (ch) {
            case 'I': return 1;
            case 'V': return 5;
            case 'X': return 10;
            case 'L': return 50;
            case 'C': return 100;
            case 'D': return 500;
            case 'M': return 1000;
            default: return 0;
        }
    }
}

```

Main.java

```

import number.Roman;
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
    }
}

```

```
        System.out.print("Enter Roman numeral: ");
        String roman = sc.nextLine();

        int result = Roman.romanToInteger(roman);

        System.out.println("Integer value: " + result);

        sc.close();
    }
}
```

Compilation and Execution

```
javac -d . number/Roman.java Main.java
java Main
```

Output

```
Enter Roman numeral: MCMXCIV
Integer value: 1994
```